ANKARA UNIVERSITY
COMPUTER ENGINEERING

REAL TIME IMAGE PROCESSING SMART SECURITY SYSTEM

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Ankara date

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1. INTRODUCTION

1.1. Problem Definition

- Huge amount of video data from surveillance cameras
- Human operators must watch screens continuously
- Operator tiredness → delayed reaction & human error
- Important events are difficult to search later
- Redundant storage of meaningless video frames

1. INTRODUCTION

1.2. Aim of the Project

- Develop an intelligent real-time surveillance assistant
- Reduce dependence on human operators
- Automatically detect security-relevant events
- Store high-quality evidence frames
- Enable fast query-based search
- Provide real-time blacklist alerts

1. INTRODUCTION

1.3. Where Can This System Be Used?

- Campus / dormitory / apartment security monitoring
- Suspicious person search: *“person wearing a red shirt”*
- Vehicle surveillance: *“blue car detected again”*
- Pet monitoring (cat/dog detection)

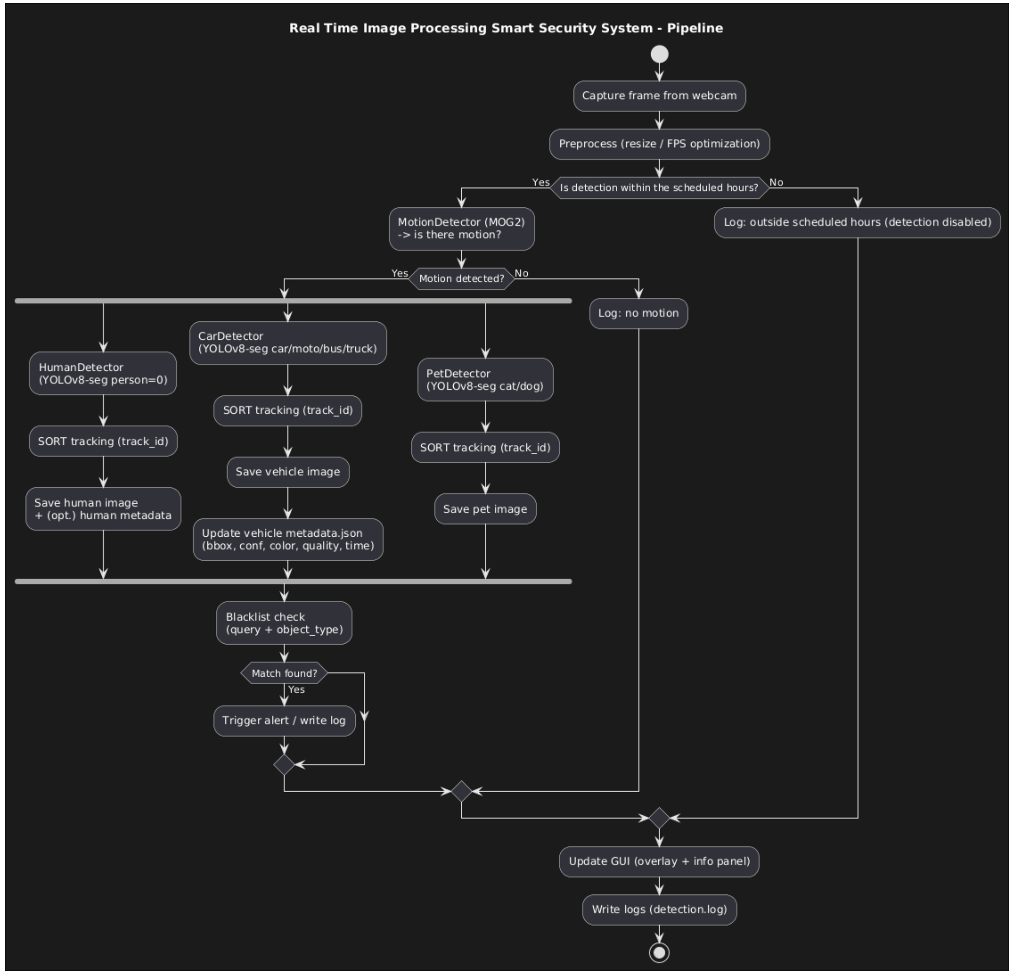
2. LITERATURE

- "*A Review on Role of Image Processing Techniques to Enhancing Security of IoT Applications*"
  - The report reviews how image-processing techniques are used to enhance the security of IoT applications, including monitoring and intrusion detection, which conceptually relates to our project’s intelligent video-based security system.
- "*Object Tracking by Detection using YOLO and SORT* "
  - The primary goal of this project is to implement real-time multi-object detection and tracking in video sequences using YOLO and SORT algorithms.
- "*Improved Adaptive Gaussian Mixture Model for Background Subtraction*" and "*Real-Time intrusion Detecting and alert system by image processing techniques* "
  - Develop a more robust and adaptive method for separating moving foreground objects from a dynamic background in video sequences, especially under changing illumination and scene variations. Motion detection was performed using a Gaussian Mixture Model–based MOG2 background subtraction approach.

3. MATERIAL & METHOD

- YOLOv8 instance segmentation
- SORT online tracking
- Background subtraction for motion detection
- HSV thresholding + temporal flickering for fire detection
- K-means clustering for color estimation
- Best-frame quality scoring
- Time-based scheduling
- Logging & event recording

4. SYSTEM ARCHITECTURE



1. Input video stream (Webcam or video file)
2. Motion detection gating
3. YOLOv8-seg inference (detection + segmentation masks)
4. SORT tracking → assigns **unique Track IDs**
5. Attribute extraction:
   Color detection from masks
   Object type/vehicle class
6. Save outputs:
   Best image per track
   JSON metadata
7. Blacklist matching:
   Notification

5. EXPERIMENTS

5.1. Experimental Design and Test Scenarios

Test environments

- Indoor classroom and corridor environments
- Outdoor daylight scenes
- Webcam stream and prerecorded video files

Evaluation components

- Motion detection accuracy
- Human, vehicle, pet detection capability
- Tracking stability (ID consistency over time)
- Fire region heuristic detection
- Evidence saving and metadata indexing
- Query-based search functionality

5. EXPERIMENTS

5.2. Experimental Results: Achievements

What worked successfully

- Real-time processing with acceptable latency
- Consistent object tracking in non-crowded scenes
- Best-frame selection significantly improved evidence clarity
- Clothing and vehicle color estimation broadly consistent under stable lighting
- Search module successfully retrieved events using color/type keywords
- Blacklist alerts triggered reliably on matching attributes

5. EXPERIMENTS

5.3. Experimental Results: Limitations

Partially achieved / needs improvement

- Color estimation affected by shadows and strong reflections
- Fire detection produced false positives in LED/bright orange scenes
- Metadata search supports simple keywords, not complex natural language

6. CONCLUSION & FUTURE WORK

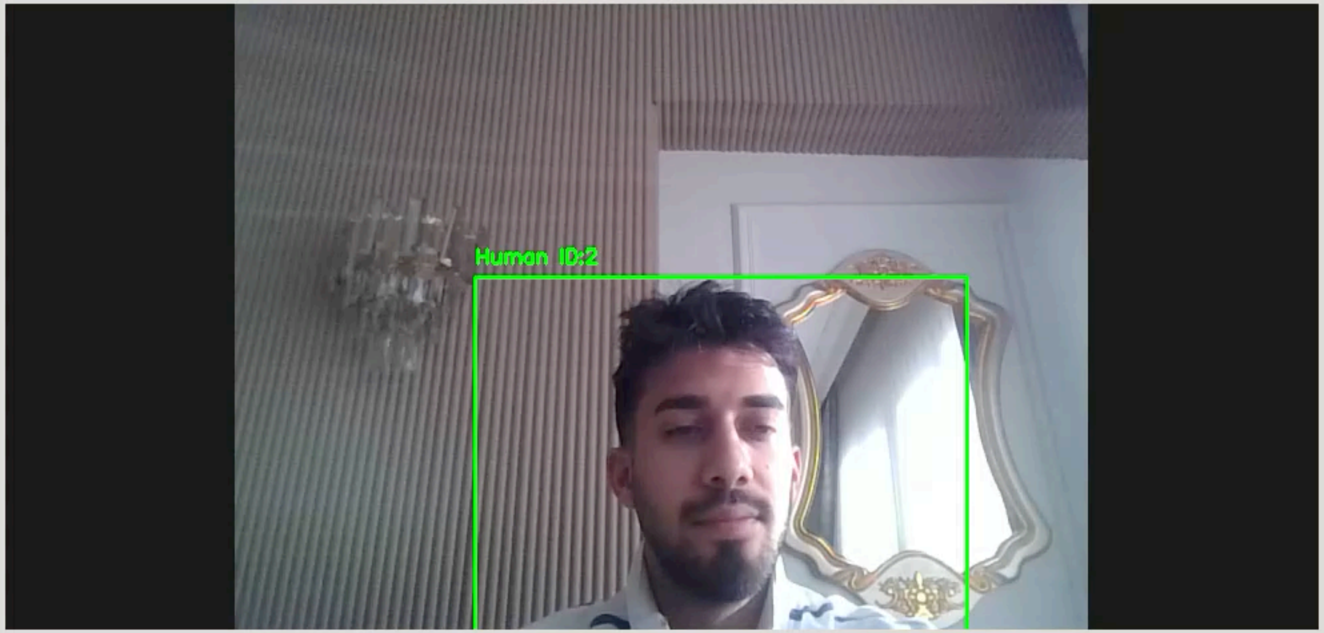
- A real-time smart security monitoring system was developed integrating motion detection, object detection, tracking, fire heuristics, evidence saving, and searchable metadata within a unified framework
- The system automatically captures the most representative evidence frames, supports query-based retrieval, and generates blacklist alerts, reducing dependency on continuous human monitoring
- Experimental results show that best-frame selection improves evidence quality and motion-gated detection increases processing efficiency in real-time scenarios
- Future enhancements will focus on improving robustness under challenging lighting and crowded scenes, adopting deep-learning-based fire detection, extending the architecture to multi-camera and edge-based deployments, enriching metadata search capabilities, and supporting mobile/remote notification mechanisms

7. DEMO

Security Camera - Human & Fire Detection System

Security Camera Monitoring System

Camera View



Controls

Status: Status: Running (camera)

Fire Detection: ☒ Enabled

Car Detection: ☒ Enabled

Pet Detection ( ): ☒ Enabled

Video Source

Webcam | Load Video | Pause/Resum

Detection Hours

Start: 0 **End:** 23 Update

Human Search

Search Query: e.g., red hat, blue car, red truck...

Time Range: Start: 0 End: 23 Search

Search Results

Security Blacklist

Description: Type: human + Add

[HUMAN] white shirt

Remove Selected

System Logs

```
[2026-01-14 16:25:32] ▶  Started on camera
[2026-01-14 16:25:48]    Saved: human_20260114_162547_id1.jpg
```

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THANK YOU

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